

Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

Group ID: _____



Annotate, mark-up, and write on this page as needed.

Round 1:

For this design exercise, you will be acting as a member of your assigned working group – you will have differing objectives and be provided performance space plots that DO NOT have any cross-disciplinary objective pairings.

INSTRUCTIONS:

Working in your assigned working group, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. You wish to maximize the vehicle's performance in your own objectives, and the other working group wishes to maximize their own objectives, but you must select a single vehicle from the 20 designs given in the supplemental packet. Write the selected design's ID in the space provided in the supplemental packet.

The Suspension and Driveline working group wishes to maximize the following objectives:

- A. Acceleration
- B. Towing Capacity

13, 20, 15

The Chassis and Cabin working group wishes to maximize the following objectives:

- C. Low Speed Maneuverability
- D. Passenger Cabin Space

20, 11, 16

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

20	13	11							
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Do you think you could have found a better solution given more time?

No, we think we were able to come to a good solution

What, in this round, stopped you from finding a better solution?

Units on the graph to gauge trade-offs

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Round 2:

For this design exercise, you will be acting as a member of your assigned working group, and you will be provided cross-disciplinary performance space plots.

INSTRUCTIONS:

Working in your assigned working group, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. You wish to maximize the vehicle's performance in your own objectives, and the other working group wishes to maximize their own objectives, but you must select a single vehicle from the 20 designs given in the supplemental packet. Write the selected design's ID in the space provided in the supplemental packet.

The Suspension and Driveline working group wishes to maximize the following objectives:

E. Top Speed

F. Off Road Ability

2,5,3

The Chassis and Cabin working group wishes to maximize the following objectives:

G. Occupant Protection

H. Maintainability

19,2,8

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

2	19	4 ↔ 11						
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Do you think you could have found a better solution given more time?

No, all good

What, in this round, stopped you from finding a better solution?

Units would be helpful. Having more graphics is nice.

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables

14	11	2	19			5			
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Do you think you could have found a better solution given more time?

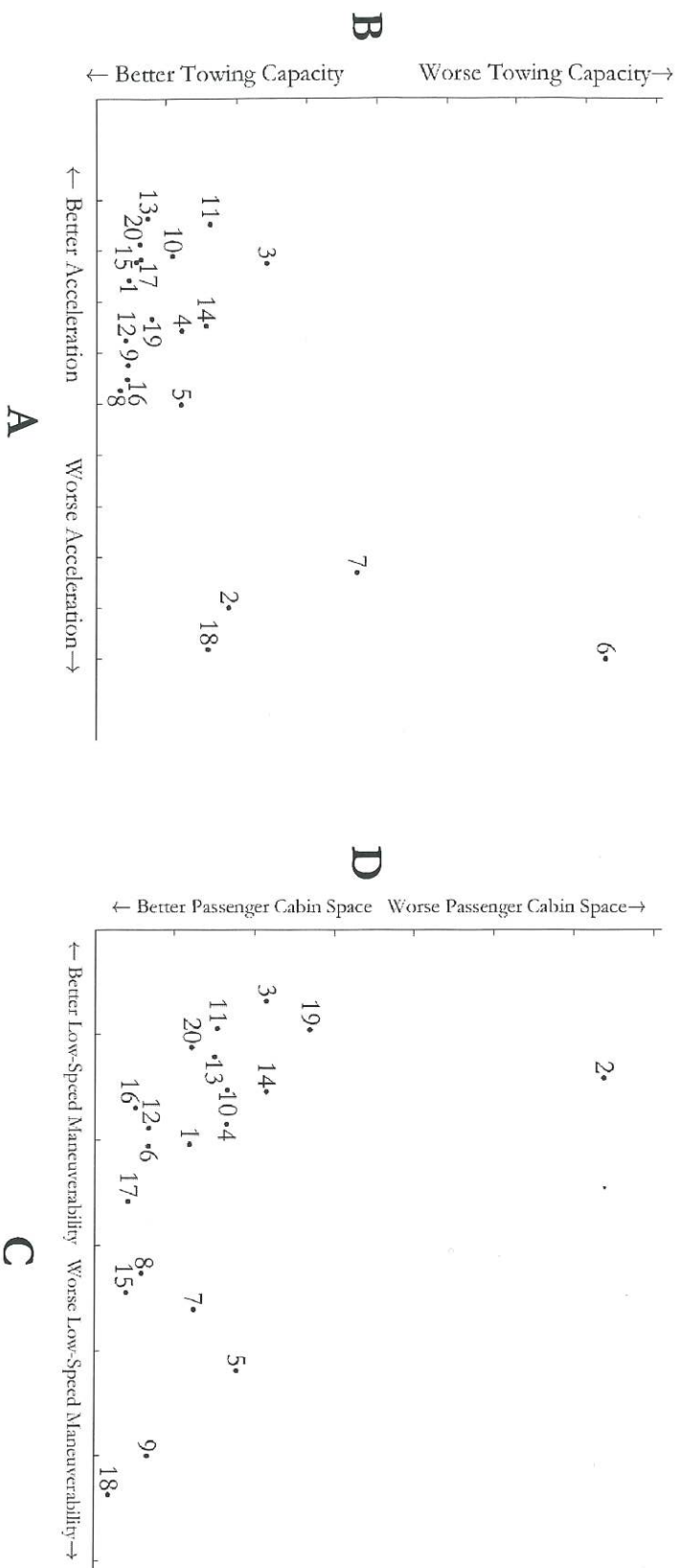
Yes, having more data (units) would be helpful.

What, in this round, stopped you from finding a better solution?

Trying to determine if three great and one awful is better or worse than four okay.

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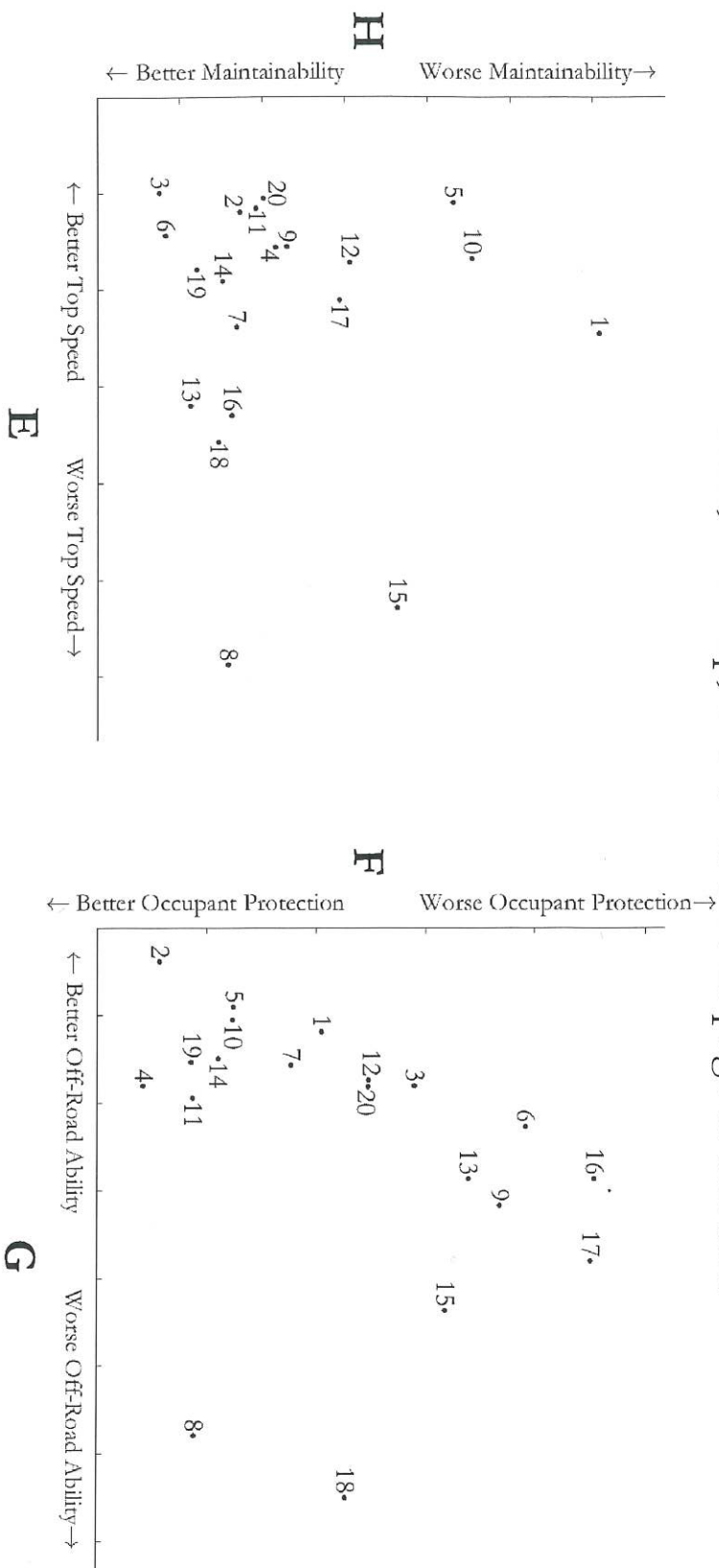
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Enter the number ID of the selected design: _____

Write your Group ID: _____

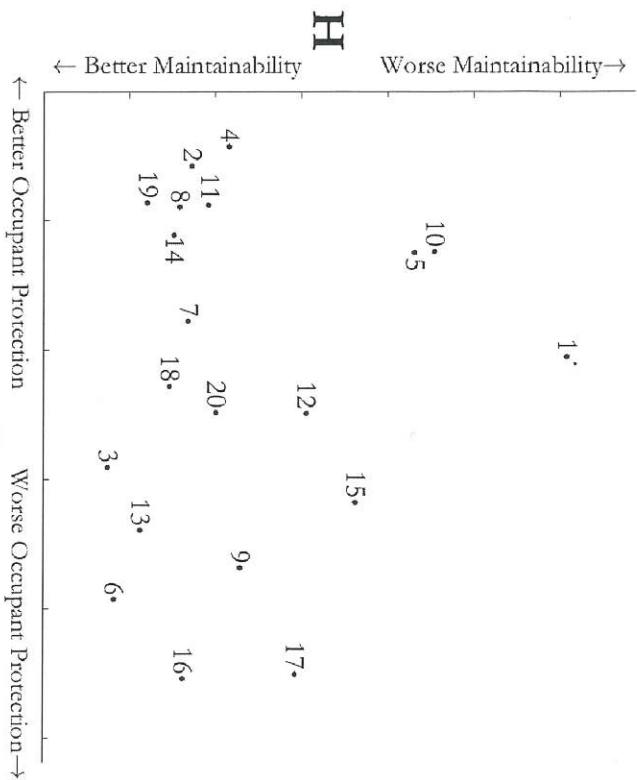
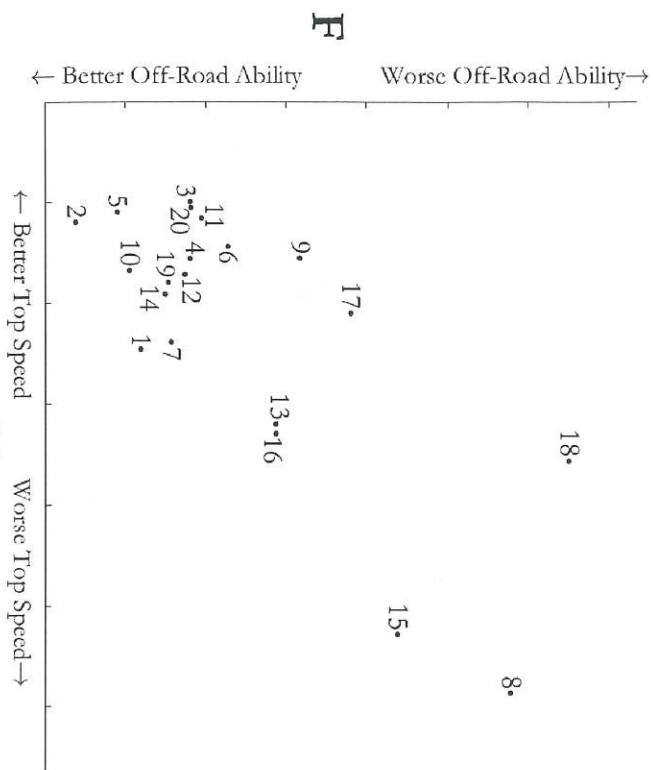
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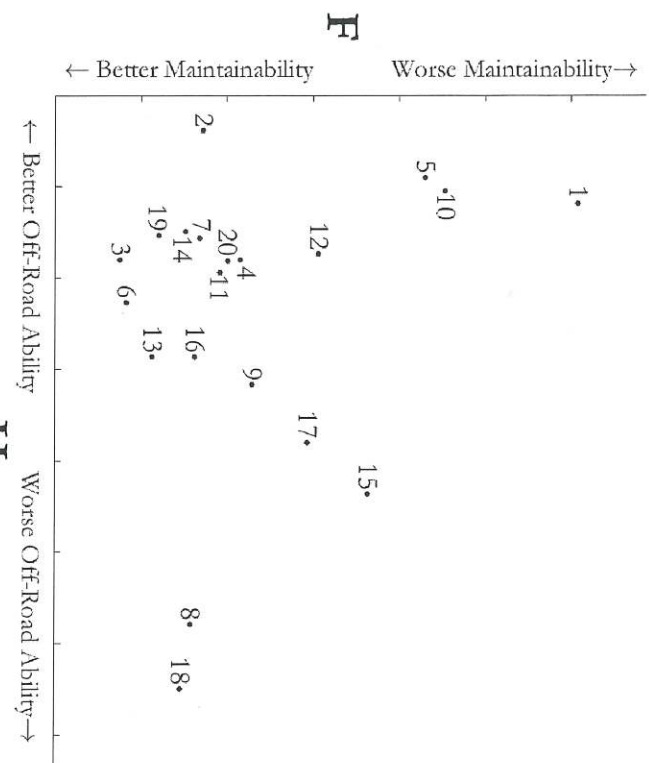
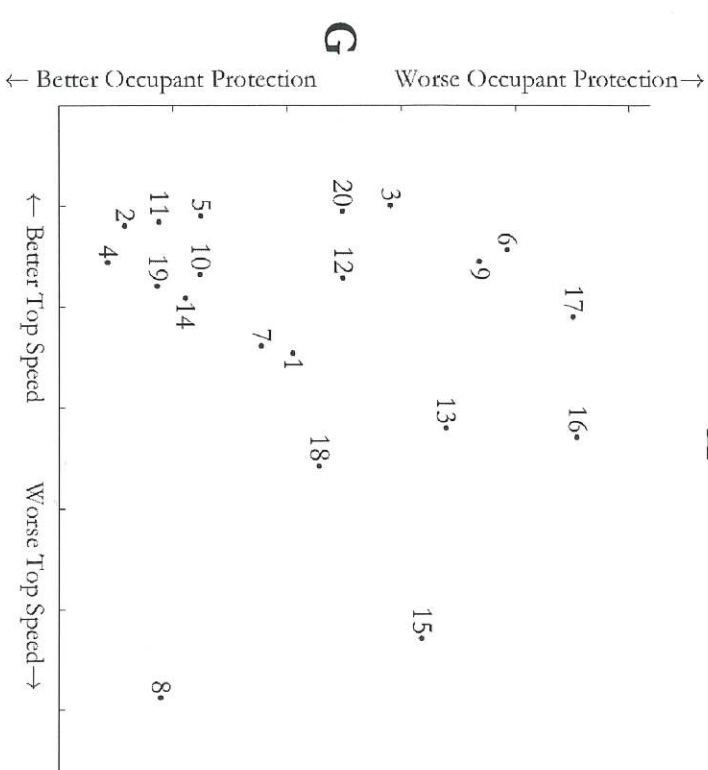
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E

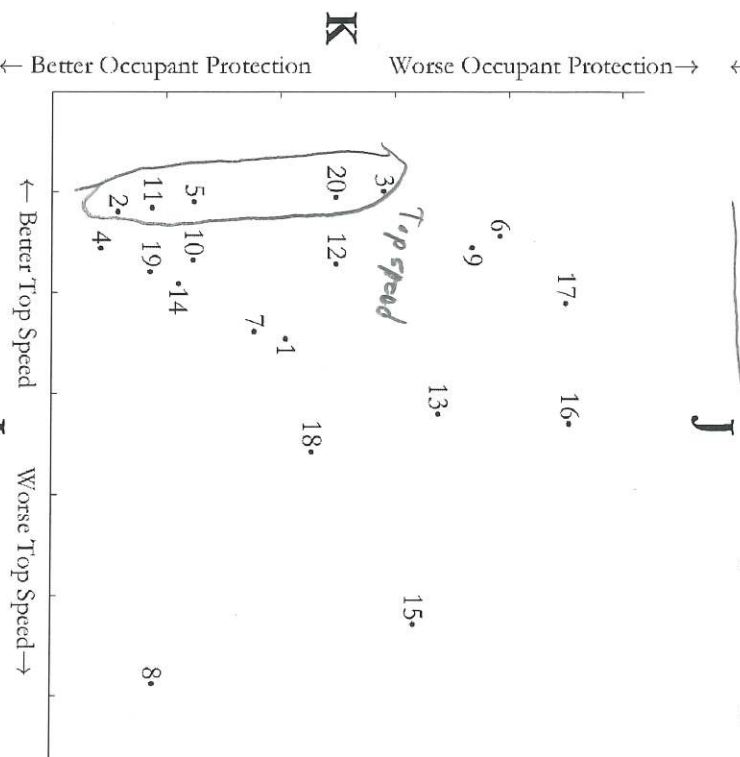
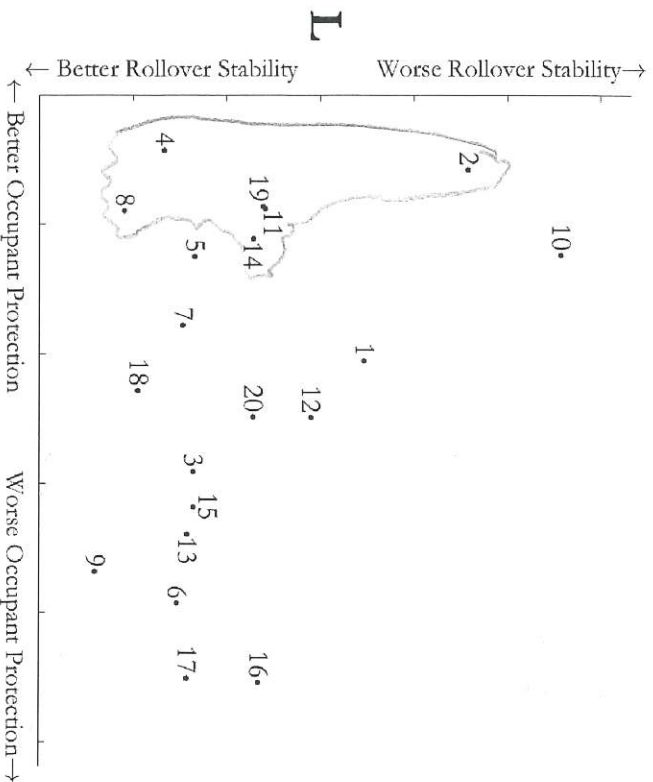
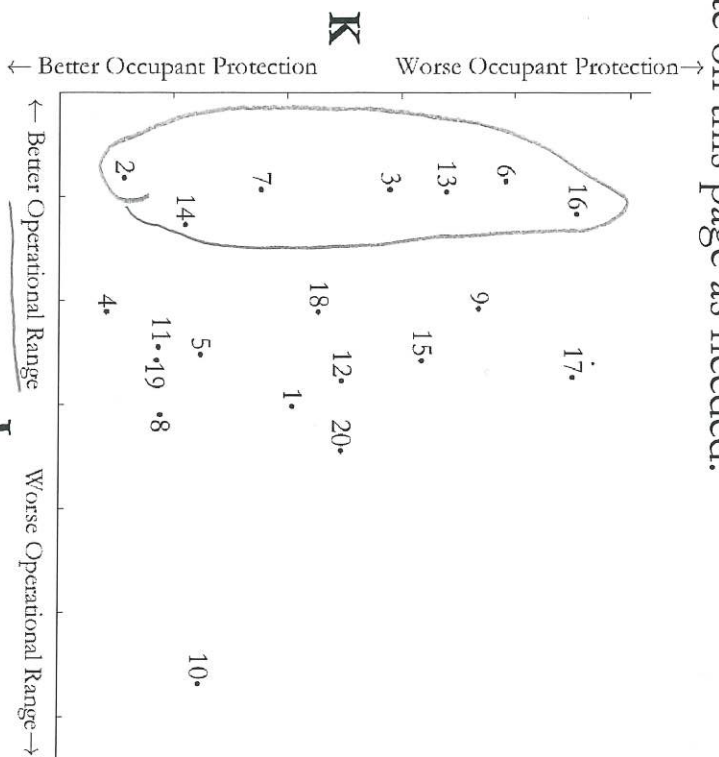
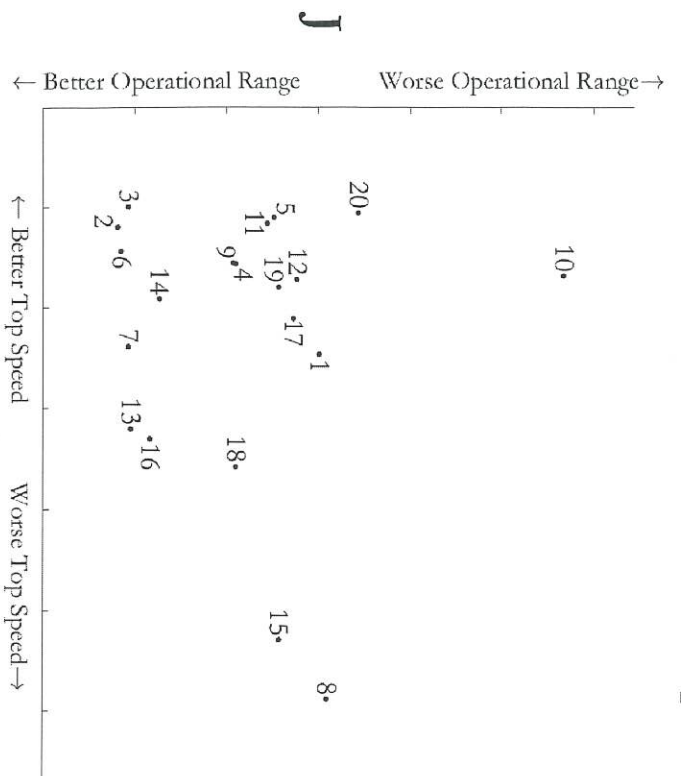
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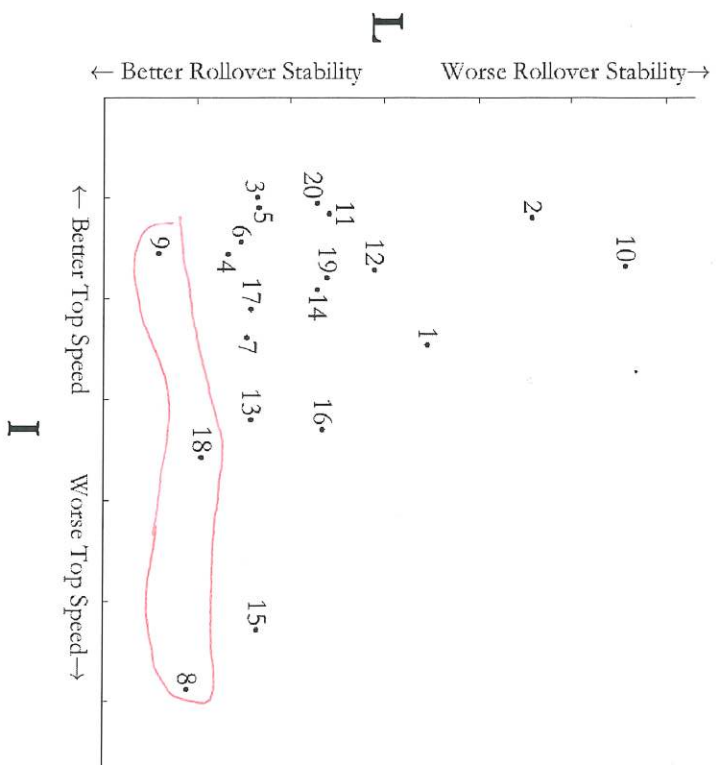
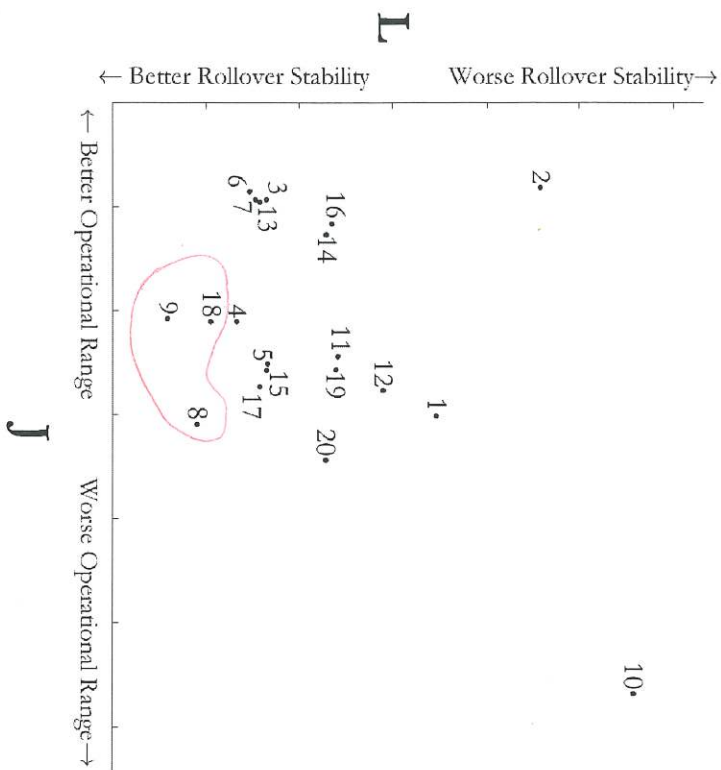
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